Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 18: Radiation

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How to generate EM waves?

- By localized systems of time-dependent charge and current densities.
- e.g., monochromatic sinusoidal oscillations.
- With Fourier transform, any source can be built.

$$\begin{cases} \nabla \cdot \vec{D} = \dfrac{\rho}{\varepsilon_0} \\ \nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t} \\ \nabla \cdot \vec{B} = 0 \\ \nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \dfrac{\partial \vec{E}}{\partial t} \end{cases}$$

Inhomogeneous Wave Equations

$$\nabla \times (\nabla \times \vec{B}) = -\frac{\partial}{\partial t}(\nabla \times \vec{B})$$

$$\Rightarrow \quad \nabla(\nabla \cdot \vec{B}) - \nabla^2 \vec{B} = -\mu_0 \frac{\partial \vec{J}}{\partial t} - \mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

$$\Rightarrow \quad \frac{\nabla \rho}{\varepsilon_0} - \nabla^2 \vec{B} = -\mu_0 \frac{\partial \vec{J}}{\partial t} - \mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}$$

$$\nabla^2 \vec{B} - \mu_0 \varepsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} = \frac{\nabla \rho}{\varepsilon_0} + \mu_0 \frac{\partial \vec{J}}{\partial t}$$

$$\begin{cases} \\ \nabla^2 \vec{B} - \mu_0 \varepsilon_0 \dfrac{\partial^2 \vec{B}}{\partial t} = -\mu_0 \nabla \times \vec{J} \end{cases}$$

Inhomogeneous Wave Equations for φ, $\vec{A}$

$$\vec{E} = -\nabla\varphi - \frac{\partial \vec{A}}{\partial t} \qquad \vec{B} = \nabla \times \vec{A}$$

Lorentz gauge: $\nabla \cdot \vec{A} + \frac{1}{c^2}\frac{\partial \varphi}{\partial t} = 0$

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \Rightarrow \quad \nabla \cdot \left(-\nabla\varphi - \frac{\partial \vec{A}}{\partial t}\right) = \frac{\rho}{\varepsilon_0}$$

$$\Rightarrow \quad -\nabla^2\varphi - \frac{\partial}{\partial t}(\nabla \cdot \vec{A}) = \frac{\rho}{\varepsilon_0}$$

$$-\nabla^2\varphi + \frac{1}{c^2}\frac{\partial^2 \varphi}{\partial t^2} = \frac{\rho}{\varepsilon_0}$$

$$\Rightarrow \quad \nabla^2\varphi - \frac{1}{c^2}\frac{\partial^2 \varphi}{\partial t^2} = -\frac{\rho}{\varepsilon_0}$$

How to solve this?

Inhomogeneous Wave Equations

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla \times (\nabla \times \vec{A}) = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$\nabla (\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$-\frac{1}{c^2} \frac{\partial}{\partial t} \nabla \varphi - \nabla^2 \vec{A} = \mu_0 \vec{J} + \frac{1}{c^2} \frac{\partial}{\partial t} \left(-\nabla \varphi - \frac{\partial \vec{A}}{\partial t} \right)$$

$$\nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu_0 \vec{J}$$

How to solve the Inhomogeneous Wave Equations?

$$\nabla^2\varphi - \frac{1}{c^2}\frac{\partial^2\varphi}{\partial t^2} = -\frac{\rho}{\varepsilon_0}$$

- Step 1: Fourier integral

$$\varphi(\vec{x},t) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} h(\vec{x},\omega)\, e^{-i\omega t}\, d\omega \quad ①$$

$$h(\vec{x},\omega) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \varphi(\vec{x},t)\, e^{i\omega t}\, dt \quad ②$$

$$\rho(\vec{x},t) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} l(\vec{x},\omega)\, e^{-i\omega t}\, d\omega \quad ③$$

$$l(\vec{x},\omega) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \rho(\vec{x},t)\, e^{i\omega t}\, dt \quad ④$$

How to solve the Inhomogeneous Wave Equations?

- Step 1: Fourier integral

① ③ $\nabla^2\varphi - \frac{1}{c^2}\frac{\partial^2\varphi}{\partial t^2} = -\frac{\rho}{\varepsilon_0}$

$\Rightarrow \quad \nabla^2 h + \frac{\omega^2}{c^2} h = -\frac{L}{\varepsilon_0}$

$\omega = kc \Rightarrow \nabla^2 h + k^2 h = -\frac{L}{\varepsilon_0}$?

We need to find the Green Function which satisfies $\nabla^2 G_k + k^2 G_k = -4\pi\delta(\vec{x}-\vec{x}')$

No boundry condition.

How to solve the Inhomogeneous Wave Equations?

- Step 1: Fourier integral

$G_k(\vec{x}, \vec{x}')$ Because no boundry condition.

G_k should be spherical symmetric.

$$G_k(\vec{x}, \vec{x}') = G_k(|\vec{x} - \vec{x}'|)$$

Define $\vec{R} = \vec{x} - \vec{x}'$ $\quad R = |\vec{x} - \vec{x}'|$

$$G_k = G_k(R)$$

How to solve the Inhomogeneous Wave Equations?

- Step 1: Fourier integral

$$\frac{1}{R}\frac{d^2}{dR^2}(RG_k) + k^2 G_k = -4\pi\delta(\vec{R})$$

When $\vec{x} \neq \vec{x}'$, $\frac{1}{R}\frac{d^2}{dR^2}(RG_k) + k^2 G_k = 0$

$$\frac{d^2}{dR^2}(RG_k) + k^2(RG_k) = 0$$

$$RG_k = Ae^{ikR} + Be^{-ikR}$$

How to solve the Inhomogeneous Wave Equations?

- Step 1: Fourier integral

$$G_k^{\pm} = \frac{e^{\pm ikR}}{R}$$

We are interested in outgoing waves

$$G_k^{+}.$$

How to solve the Inhomogeneous Wave Equations?

- Step 2: Green's theorem

$$\int (\phi \nabla^2 \psi - \psi \nabla^2 \phi)\, dV = \int_S \left[\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n}\right] dS$$

Let $\phi = h$, $\psi = G_k^+$

Assume RHS = 0 (provable)

$$\Rightarrow \quad -4\pi h + \int G_k^+ \frac{L}{\varepsilon_0} = 0$$

$$\Rightarrow \quad h(\vec{x}, w) = \frac{1}{4\pi\varepsilon_0} \int L \frac{e^{ikR}}{R} dV'$$

Retarded Potential

$$\varphi(\vec{x},t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{4\pi\varepsilon_0} \int \frac{L e^{ikR}}{R} dv' e^{-i\omega t} d\omega$$

$$= \frac{1}{4\pi\varepsilon_0} \frac{1}{\sqrt{2\pi}} \iint \frac{e^{ikR - i\omega t}}{R} L \, d\omega dv'$$

$$\rho(\vec{x},t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} L(\vec{x},\omega) e^{-i\omega t} d\omega$$

$$\rho\left(\vec{x}, t - \frac{R}{c}\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} L(\vec{x},\omega) e^{-i\omega t + ikR} d\omega$$

Retarded Potential

$$\Phi(\vec{x},t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{4\pi\varepsilon_0} \int \frac{\rho e^{ikR}}{R} \, dV' \, e^{-i\omega t} \, d\omega$$

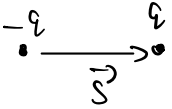
$$= \frac{1}{4\pi\varepsilon_0} \frac{1}{\sqrt{2\pi}} \iint \frac{e^{ikR - i\omega t}}{R} \rho \, d\omega \, dV'$$

$$= \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\vec{x}, t - \frac{R}{c})}{R} \, dV' \qquad \rho_{ret} = \rho(\vec{x}, t - \frac{R}{c})$$

$$\vec{A} = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{x}, t - \frac{R}{c})}{R} \, dV' \qquad \vec{J}_{ret} = \vec{J}(\vec{x}, t - \frac{R}{c})$$

Example: Time-Dependent Electric Dipole

- What is the fields generated by a monochromatic sinusoidal electric dipole?
- Step 1: Find the source



$$\vec{p}(t) = q\vec{s}\,e^{-i\omega t}$$

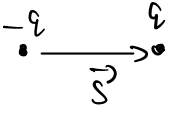
$$\varphi = \lim_{|\vec{s}|\to 0} \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{|\vec{r}-\vec{s}|} - \frac{q}{|\vec{r}|}\right)$$

when $|\vec{s}|\to 0$, $\frac{1}{|\vec{r}-\vec{s}|} = \frac{1}{|\vec{r}|} + \vec{s}\cdot\nabla\frac{1}{|\vec{r}|} + \cdots$

$$\varphi = \frac{1}{4\pi\varepsilon_0}\vec{p}\cdot\nabla\frac{1}{|\vec{r}|}$$

Example: Time-Dependent Electric Dipole

- What is the fields generated by a monochromatic sinusoidal electric dipole?
- Step 1: Find the source



$$\nabla^2\varphi = -\frac{\rho}{\varepsilon_0} \quad \Rightarrow \quad \rho = -\varepsilon_0\nabla^2\varphi$$

$$\rho = -\varepsilon_0\nabla^2\left(\frac{1}{4\pi\varepsilon_0}\vec{p}\cdot\nabla\frac{1}{|\vec{r}|}\right)$$

$$= -\frac{1}{4\pi}\vec{p}\cdot\nabla\left(\nabla^2\frac{1}{|\vec{r}|}\right) = -\frac{1}{4\pi}\vec{p}\cdot\nabla\left(4\pi\delta(|\vec{r}|)\right)$$

$$= -\nabla\cdot\left(\vec{p}\,\delta(|\vec{r}|)\right)$$

Example: Time-Dependent Electric Dipole

- What is the fields generated by a monochromatic sinusoidal electric dipole?
- Step 1: Find the source

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0$$

$$\frac{\partial}{\partial t}\left(-\nabla \cdot \left(\vec{P}\delta(|\vec{r}|)\right)\right) + \nabla \cdot \vec{J} = 0$$

$$\vec{J}(\vec{x}, t) = \delta(\vec{r})\frac{d\vec{P}(t)}{dt} = \delta(\vec{r})\dot{\vec{P}}(t)$$

Example: Time-Dependent Electric Dipole

- Step 2: Find the vector potential

Example: Time-Dependent Electric Dipole

- Step 3: Find the scalar potential

Example: Time-Dependent Electric Dipole

Find the fields

$$\mathbf{B}(\mathbf{r},t) = -\frac{\mu_0}{4\pi}\hat{\mathbf{r}} \times \left\{ \frac{\dot{\mathbf{p}}_{\text{ret}}}{r^2} + \frac{\ddot{\mathbf{p}}_{\text{ret}}}{cr} \right\}$$

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0} \left\{ \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{p}_{\text{ret}}) - \mathbf{p}_{\text{ret}}}{r^3} + \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \dot{\mathbf{p}}_{\text{ret}}) - \dot{\mathbf{p}}_{\text{ret}}}{cr^2} + \frac{\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}}_{\text{ret}}) - \ddot{\mathbf{p}}_{\text{ret}}}{c^2 r} \right\}.$$

- Near zone

Example: Time-Dependent Electric Dipole

$$\mathbf{B}(\mathbf{r},t) = -\frac{\mu_0}{4\pi}\hat{\mathbf{r}} \times \left\{ \frac{\dot{\mathbf{p}}_{\text{ret}}}{r^2} + \frac{\ddot{\mathbf{p}}_{\text{ret}}}{cr} \right\}$$

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0} \left\{ \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{p}_{\text{ret}}) - \mathbf{p}_{\text{ret}}}{r^3} + \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \dot{\mathbf{p}}_{\text{ret}}) - \dot{\mathbf{p}}_{\text{ret}}}{cr^2} + \frac{\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}}_{\text{ret}}) - \ddot{\mathbf{p}}_{\text{ret}}}{c^2 r} \right\}.$$

- Intermediate zone

Example: Time-Dependent Electric Dipole

$$\mathbf{B}(\mathbf{r},t) = -\frac{\mu_0}{4\pi}\hat{\mathbf{r}} \times \left\{ \frac{\dot{\mathbf{p}}_{\text{ret}}}{r^2} + \frac{\ddot{\mathbf{p}}_{\text{ret}}}{cr} \right\}$$

$$\mathbf{E}(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0} \left\{ \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{p}_{\text{ret}}) - \mathbf{p}_{\text{ret}}}{r^3} + \frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \dot{\mathbf{p}}_{\text{ret}}) - \dot{\mathbf{p}}_{\text{ret}}}{cr^2} + \frac{\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \ddot{\mathbf{p}}_{\text{ret}}) - \ddot{\mathbf{p}}_{\text{ret}}}{c^2 r} \right\}.$$

- Far zone

Definition of Radiation

Radiation Fields

Radiation Fields

Radiation Property

Radiation Property

